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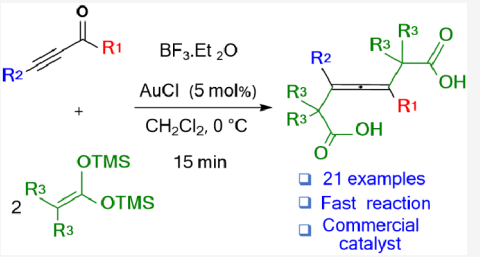
Synthesis of Functionalized Tetrasubstituted Allenes by the Addition of Bis(trimethylsilyl)ketene Acetals to Ynones Catalyzed by Gold(I)

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ABSTRACT: We have developed a straightforward and rapid methodology for the synthesis of tetrasubstituted allenes bearing carboxylic acids in the 1,3-position through the gold(I)-catalyzed nucleophilic addition of bis(trimethylsilyl)ketene acetals to ynones. The reaction was evaluated with several substrates, and 21 allenes were obtained in moderate to good yields. Using DFT calculations, we studied the mechanism of the reaction, which suggested a nucleophilic 1,4-addition pathway. The potential of allenes to act as a source of highly functionalized lactones was also explored.



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INTRODUCTION

Over the years, research on allene chemistry has continuously grown and evolved, as these compounds are highly versatile and present in various natural products, bioactive compounds, and drugs.[1] Allenes are recognized as crucial precursors to more complex organic compounds.[2] Due to their unique structure and electronic properties, they have also been object of study in coordination chemistry[3] and materials science.[4]

In recent years, several methodologies for synthesizing allenes have been developed using transition metal catalyst like palladium,[5] copper,[6] and gold.[7] In the case of gold, the transformations generally involve isomerization of propargyl acetate derivatives, rearrangement of propargyl derivatives, and S_N2 reactions.

Although the synthesis of allenes has been widely explored,[8] obtaining tetrasubstituted and highly functionalized allenes is still a recurrent challenge, in part due to their high reactivity when the substituents induce intramolecular cyclization.[9]

In previous work, when we were studying the reactivity of ynones toward the nucleophilic addition of bis(trimethylsilyl)-ketene acetals for the synthesis of 3-hydroxy-pent-4-ynoic acids, we discovered the formation of the tetrasubstituted allene **3** as a side product, exclusively when the substituent R^1 in the ynone was *p*-ClPh, *p*-MePh, and *p*-OMePh (Scheme 1).[10] Interestingly, investigations into the electronic properties of allene **3** (R^1 = *p*-OMePh, *p*-MePh) revealed semiconductor behavior when this phenomenon was investigated over thin films prepared by vapor deposition of **3** doped with tetracyano-*p*-quinodimethane (TCNQ),[4] recently taking advantage of this property, the preparation of semiconductor membranes consisting of magnesium phthalocyanine-allene **3** (*R* = *p*-OMePh) particles dispersed in nylon-11 films was explored.[4]

Encouraged by these findings, we developed a general methodology for the synthesis of tetrasubstituted allenes **3**, through the nucleophilic addition of bis(trimethylsilyl)ketene acetal **2** to ynone **1** catalyzed by AuCI.

RESULTS AND DISCUSSION

We began the study of the reaction conditions (Table 1) using ynone **1a** (1 equiv) as a model substrate, $BF_3{\cdot}Et_2O$ (1 equiv) as a Lewis acid, and bis(trimethylsilyl)ketene acetal **2a** (2 equiv) in dichloromethane as a solvent. The noncatalyzed reaction between **1a** and **2a** produced allene **3a** in a yield of only 40% after 24 h, with a recovery of about 55% of the starting material (entry 1). Thus, we started the catalytic evaluation with AuCl (5 mol %) in the absence of the Lewis acid (entry 2), but no reaction was observed, revealing the importance of both carbonyl and alkyne activation to achieve the double addition of the bis(trimethylsilyl)ketene acetal to the ynone **1a** to produce allene **3a**. Using 5 mol % of AuCl and 1 equiv of $BF_3{\cdot}Et_2O$, the starting material was completely consumed in just 15 min at 0 °C. The desired allene **3a** was then obtained in a yield of 70% (entry 3). Notably, the reaction efficiency was significantly improved under these conditions. Allenoic acid derivatives are frequently cyclo-isomerized by gold catalysts.[11] Interestingly, the gold chloride

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Scheme 1. Synthesis of Tetrasubstituted Allenes 3

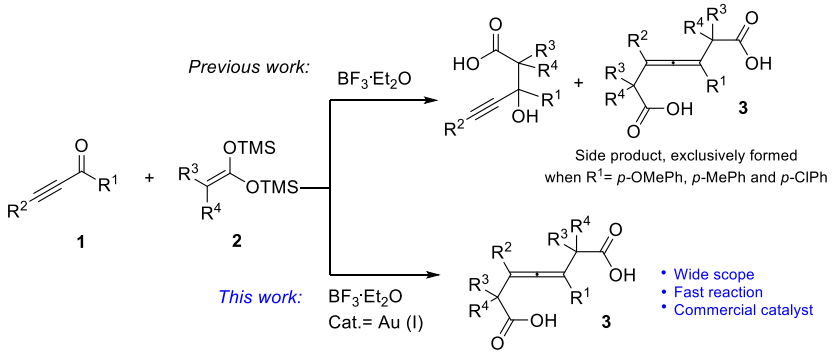
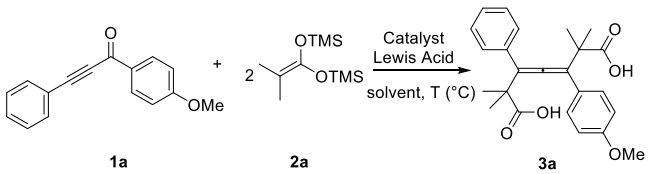


Table 1. Optimization of Reaction Conditions for the Synthesis of Allene 3a[a]



entry	catalyst	Lewis acid	solvent	T (°C)	time	yield (%)[b]
1	None	$BF_3·Et_2O$	CH_2Cl_2	0	24 h	40
2	AuCl (5 mol %)	None	CH_2Cl_2	0	24 h	n.r
3	AuCl (5 mol %)	$BF_3·Et_2O$	CH_2Cl_2	0	15 min	70
4	AuCl (3 mol %)	$BF_3·Et_2O$	CH_2Cl_2	0	15 min	51
5	AuCl (5 mol %)	$FeCl_3$	CH_2Cl_2	25	24 h	n.r
6	AuCl (5 mol %)	$BiBr_3$	CH_2Cl_2	0	10 min	23
7	CuBr (5 mol %)	$BF_3·Et_2O$	CH_2Cl_2	0	24 h	35
8	$RuCl_3·3H_2O$ (5 mol %)	$BF_3·Et_2O$	CH_2Cl_2	0	10 min	53
9	$AuCl(SC_4H_8)$ (5 mol %)	$BF_3·Et_2O$	CH_2Cl_2	0	10 min	58
10	$(Ph_3P)AuCl$ (5 mol %)	$BF_3·Et_2O$	CH_2Cl_2	0	15 min	57
11	AuCl (5 mol %)	$BF_3·Et_2O$	THF	0	24 h	28
12	AuCl (5 mol %)	$BF_3·Et_2O$	Et_2O	0	24 h	42
13	AuCl (5 mol %)	$BF_3·Et_2O$	CH_2Cl_2	25	15 min	66
14	AuCl (5 mol %)	$BF_3·Et_2O$	CH_2Cl_2	−30	1 h	5

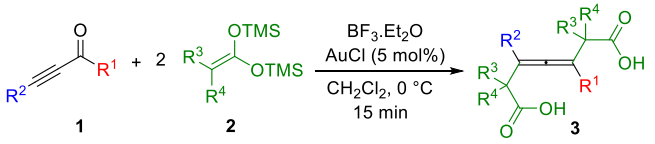
[a]Reaction conditions: **1a** (0.85 mmol), **2a** (1.87 mmol, 2.2 equiv), Lewis acid (0.85 mmol), dry solvent (15 mL), under nitrogen atmosphere. [b]Isolated yields.

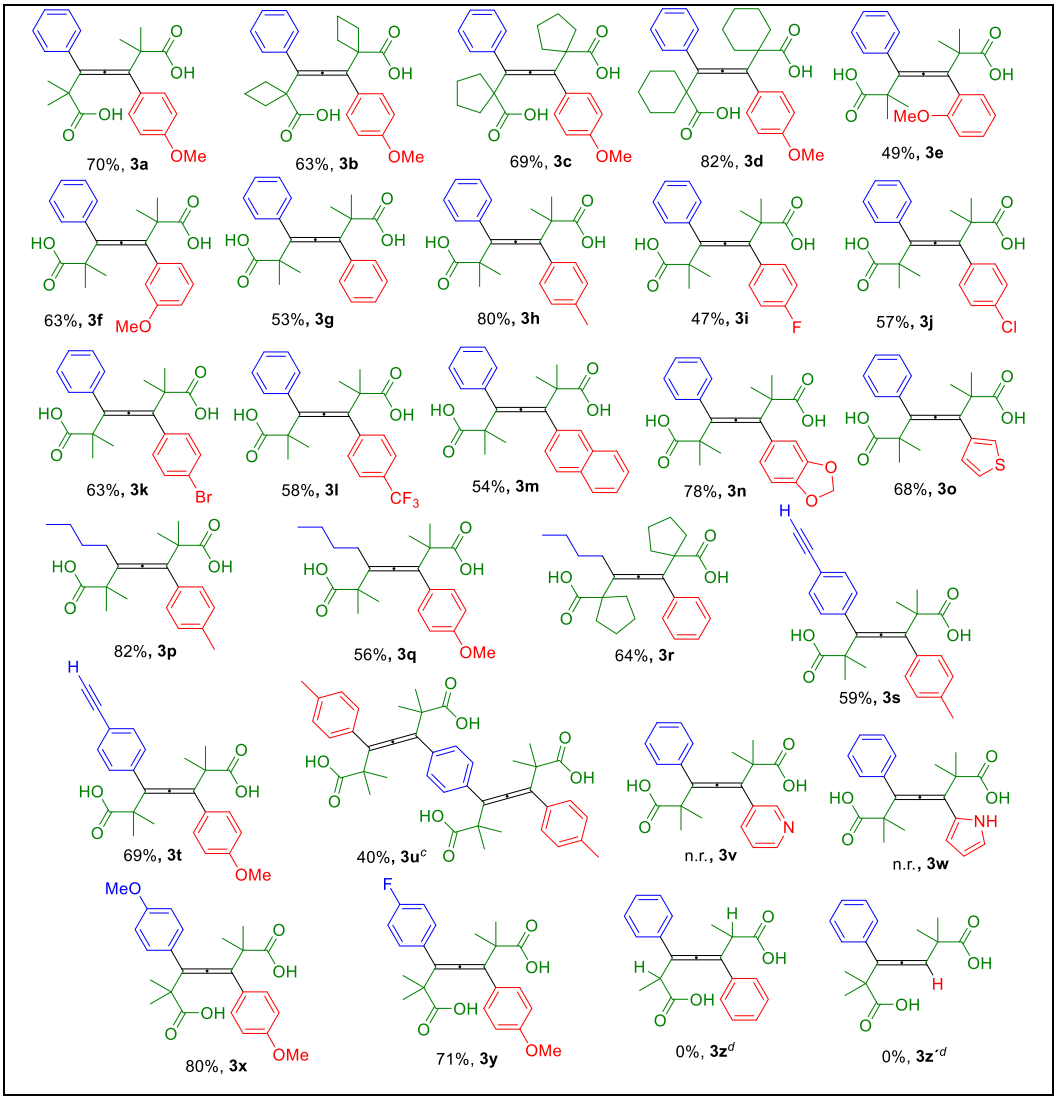
(I) catalyst did not cause further activation of allenoic acid **3a**, and no cycloisomerization product was detected. Then, we decreased the gold catalyst charge to 3 mol %, but the product was obtained in only 51% yield (entry 4). Other Lewis acids were tested for the activation of the carbonyl group. Ynone **1a** and ketene acetal **2a** did not react in the presence of $FeCl_3$ (1 equiv) (entry 5). Using $BiBr_3$ resulted in a prompt reaction, however, a complicated mixture was produced, and allene **3a** was isolated in 23% yield (entry 6). At this stage, we opted to retain $BF_3·Et_2O$ as the Lewis acid and proceeded to evaluate other potential catalysts such as $AuCl(SC_4H_8)$, $(Ph_3P)AuCl$ (5 mol %), and Cu(I) and Ru(III) commercial salts (entries 7−10), but the catalytic activity was not improved compared to that of AuCl (entry 3). It was also observed that the reaction yield decreases in the presence of coordinating solvents like tetrahydrofuran and diethyl ether (entries 11 and 12). The reactions described utilizing $BF_3·Et_2O$ were executed at a temperature of 0 °C to prevent its volatilization. Other temperatures were also investigated. At 25 °C, a decrease in yield was observed (entry 13), which was attributed to the release of the volatile $BF_3·Et_2O$. Meanwhile, at −30 °C, the reaction was too slow, yielding only 5% of **3a** after 1 h (entry 14).

To investigate the scope of the reaction, we applied the best reaction conditions stated in entry 3, Table 1. First, we focused on modifying the substituent of bis(trimethylsilyl)ketene acetal R^3, R^4. By introducing cyclobutyl, cyclopentyl, and cyclohexyl, we achieved successful synthesis of allenes **3b−3d** (Table 2), with a higher yield observed for less strained ring **3d**. Then, the effect of substituent R^1 in the α-position of the ketone was examined. A decrease in reaction efficiency was noted when the methoxyl group was changed from the *para* to *meta* and *ortho* positions (**3e**, **3f**), with a significantly lower yield of the ortho-substituted derivative **3e**. The structure of **3e** was unambiguously established by an X-ray crystallography analysis. The incorporation of different electron-donating and electron-withdrawing groups in the *para* position of phenyl group R^1 was well tolerated, producing allenes **3** from moderate to good yields (**3g−3l**), standing out the best results when the phenyl group is substituted by electron-donating groups like methyl and methoxyl (**3a** and **3h**). Other aromatic rings, such as 1,3-benzodioxolyl, naphthyl, and 3-thienyl groups, also provided the desired products in good yields (**3m−3o**). Nevertheless, the introduction of 3-pyridyl and 2-pirrolyl R^1 substituents (**3v** and **3w**) presented a limitation in this approach. This was probably due to the preference of $BF_3·Et_2O$ to coordinate the nitrogen atom of the heterocycle rather than to activate the carbonyl group. Next, variation of alkyne substituent $\mathbf{R^2}$ was also investigated. When $\mathbf{R^2}$ is an alkyl group the reaction proceeded without any particularity affording the corresponding allenes (**3p−3r**) in good yields. Furthermore, the structure of allene **3q** was fully established by means of X-ray diffraction analysis (Figure 1). Gratefully, substrates which bear a terminal alkyne in the *para* position of the phenyl substituent $\mathbf{R^2}$ are compatible (**3s** and **3t**), and an additional activation of the terminal alkyne was not observed, which remained intact. Besides, allenes with *para* substituted phenyl groups in $\mathbf{R^2}$ like the electrondonating group methoxyl **3x** and the electronwithdrawing group fluor **3y** were achieved in good yields. A more complex structure, the diallene **3u** was

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Table 2. Scope of the Reaction for the Synthesis of Allene 3[a,b]





[a]Reaction conditions: **1** (4.23 mmol, 1 Equiv), **2a** (9.3 mmol, 2.2 Equiv), $BF_3.Et_2O$ (4.23 mmol, 1 Equiv), AuCl (5 mol %), dry CH_2Cl_2 (15 mL), under nitrogen atmosphere. [b]Isolated yields. [c]4.2 Equiv of **2a** were added. n.r = No reaction was observed. [d]Complex rection mixture, the product was not observed.

successfully synthesized in 40% yield starting from 3,3′-(1,4-phenylene)bis(1-(*p*-tolyl)prop-2-yn-1-one).

It is important to mention that this methodology has some limitations, when bis(trimethylsilyl)ketene acetal is monosubstituted R^3 = H and R^4 = Me a complex reaction mixture is produced, that is the case of allene **3z** which was not observed. When R^1 is an alkyl group or hydrogen several products are formed, in this way allene **3z′** was not possible to synthesize.

During examination of the possible reaction mechanism, two potential pathways were explored. The first pathway was triggered by the initial nucleophilic attack of the bis-(trimethylsilyl)ketene acetal via 1,4-addition, while the second was triggered by a 1,2-addition (Scheme 2). To distinguish between the two possibilities, an energy profile was computed by density functional theory (DFT) calculations using 1,3-diphenyl-2-propyn-1-one as a model substrate (Figure 2).

DFT Studies. Geometry optimizations were carried out under the tight binding approximation GFN2-xTB[12] using the XTB program. This procedure represents an efficient computational protocol for robust transition state localization that can be applied to complex reactions involving metallic centers.[13] We carried out harmonic frequency calculations for all of the computed structures to prove their nature as local minima (without imaginary frequencies) or saddle points (with exactly one imaginary frequency) for the case of transition states. Afterward, we completed single point calculations with the pw6b95 functional[14] and the def2-TZVPP basis set[15] together with atom-pairwise dispersion correction with the Becke−

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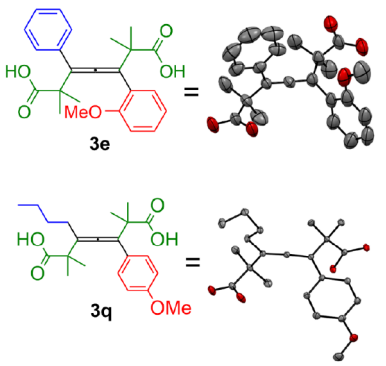


Figure 1. ORTEP representation of allenes **3e** and **3q**, ellipsoids shown at the 50% probability level for both structures.

Johnson damping scheme[16] and the SMD solvation approach.[17] All DFT calculations were completed with the help of the ORCA package[18] using a normal grid (Grid 6 option). All of the discussed energies are Gibbs free relative energies at 298.1 K and 1 atm, and they are reported in kcal/mol.

According to the DFT calculations, the reaction is proposed to work via the 1,4-addtion (Figure 2). First, ynone **1** is activated by the coordination of $BF_3{\cdot}Et_2O$ to the carbonyl group and the coordination of AuCl to the triple bond, to generate adduct **A**. Then, the nucleophilic attack of bis-(trimethylsilyl)ketene acetal to the activated alkyne in adduct **A** proceeds via transition state **TS1** with a small energy barrier, to give the intermediate **B**, which is 12.8 kcal/mol more stable than the intermediate **C** generated by the 1,2-addtion pathway. A second nucleophilic attack occurs toward the activated carbonyl group of intermediate **B**, which is the rate-limiting reaction step, leading to the formation of the intermediate **Y** via the transition state **TS2**. The energy barrier for this step was calculated to be 4.5 kcal/mol.

In contrast, for the pathway via 1,2-addition, the transformation of intermediate **C** to **Y** requires greater energy, which is calculated to be 17.9 kcal/mol. Finally, intermediate **Y** evolves through an elimination step, leading to **Z** which easily hydrolyzes to afford allene **3**.

Finally, the reactivity of allene **3n** in a cycloisomerization reaction using CuI (5 mol %) as catalyst in DMF at 100 °C under microwave irradiation is shown. *Z*-Lactone **4** was obtained as a major isomer in 80% yield (Scheme 3).

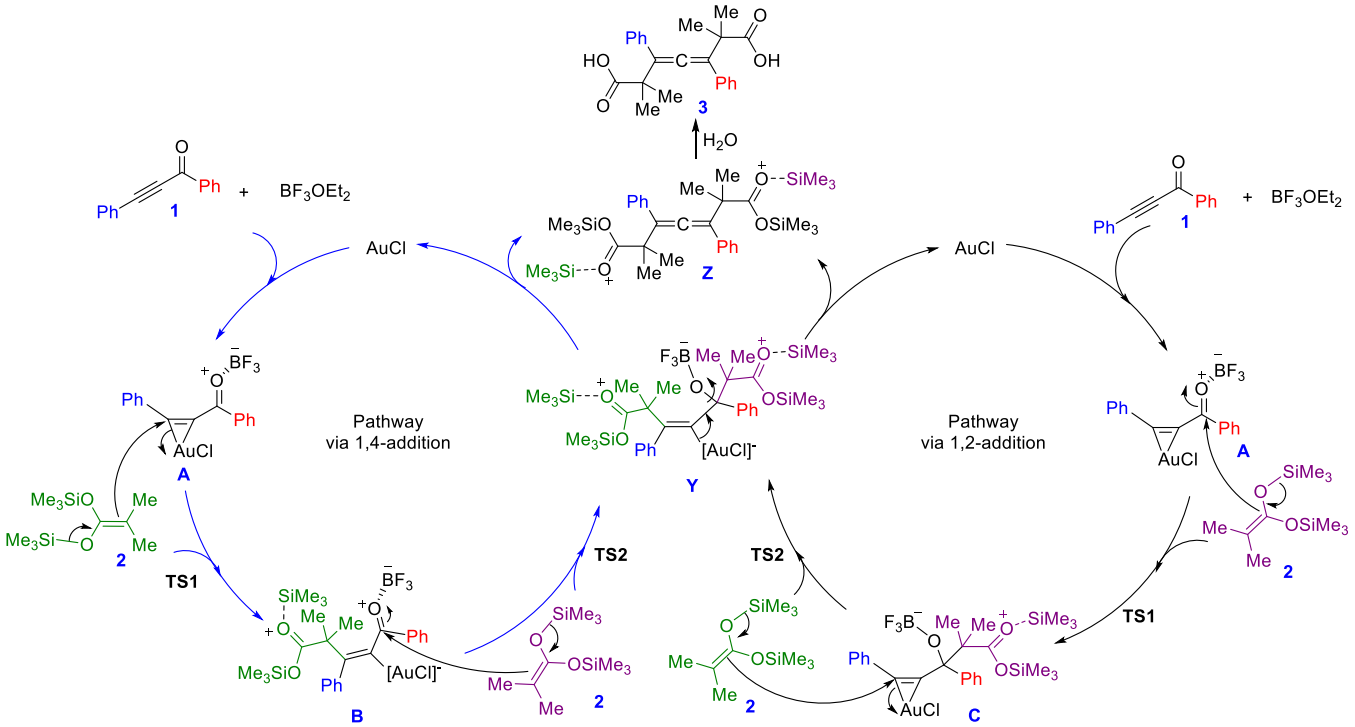
CONCLUSIONS

We developed a general and fast methodology to access highly functionalized tetrasubstituted allenes **3**, through the nucleophilic addition of bis(trimethylsilyl)ketene acetals to activated ynones by a Lewis acid and catalyzed by a commercially available gold(I) salt under mild conditions. Several ynones were tested, demonstrating a broad scope of the reaction by the production of allenes **3** from moderate to good yields, with the exception of ynones substituted by pyridyl and pyrrolyl, which did not react. The density functional theory (DFT) calculations suggested that the mechanism reaction proceeds via a nucleophilic 1,4-addition, where the first nucleophilic addition of the bis(trimethylsilyl)ketene acetal to the activated ynone readily occurs with a low energy barrier, while addition of the second molecule of bis(trimethylsilyl)ketene is the rate limiting step. The reactivity of allene **3n** was demonstrated by a cycloisomerization reaction, showing the potential of these substrates as precursors of highly functionalized lactones.

EXPERIMENTAL SECTION

General Information. All substrates and solvents were purchased from specialized suppliers with analytical purity and

Scheme 2. Plausible Mechanism for the Formation of Allene 3 via 1,4-Addition to Ynone 1



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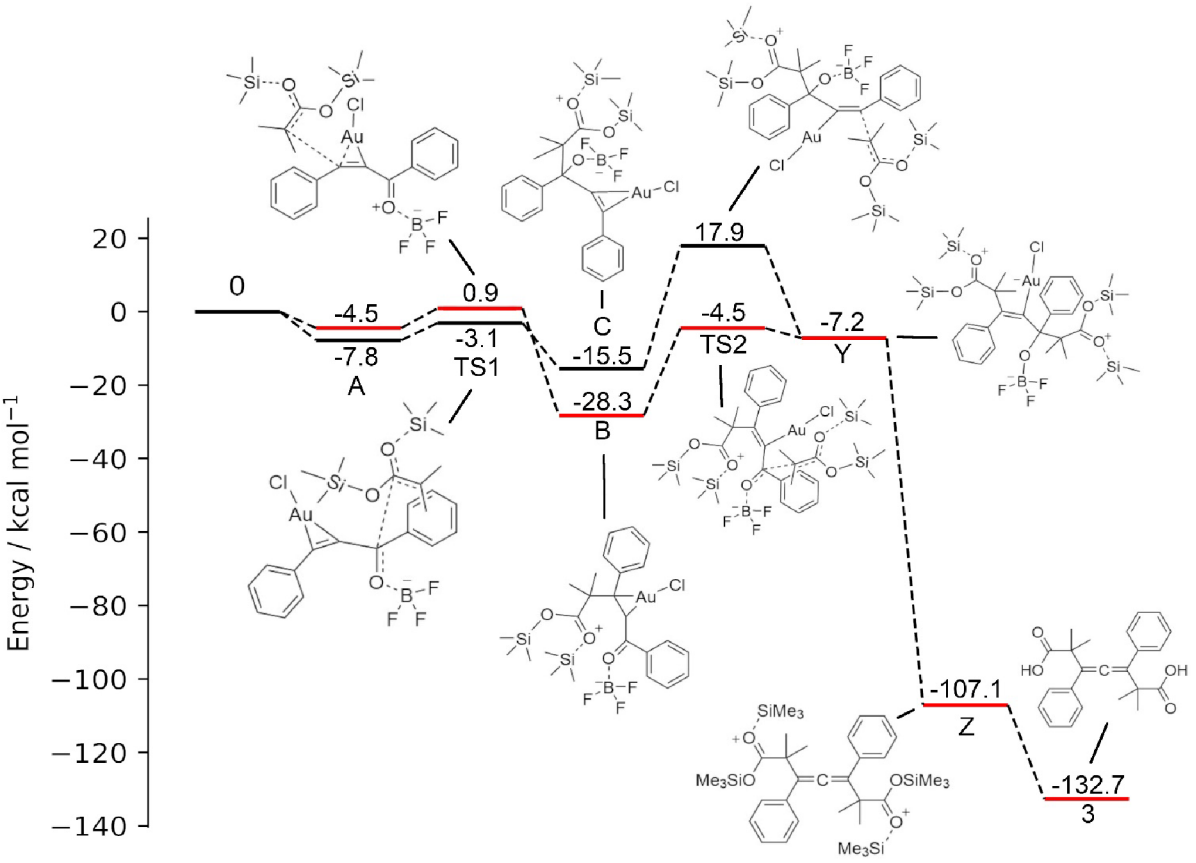
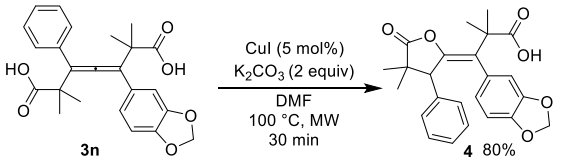


Figure 2. Energy profile of the reaction. Black: pathway via 1,2-addition. Red: pathway via 1,4-addition.

Scheme 3. Cycloisomerization Reaction of Allene 3n



were used as received without any further purification. Melting points were determined on a Melt Temp II apparatus. The ^{1}H and ^{13}C NMR spectra were measured with a Bruker Avance III (300 MHz) and an Eclipse Jeol (300 MHz) using $CDCl_3$, DMSO, and acetone as solvent. Chemical shifts are expressed in ppm (δ) relative to TMS. The following abbreviations are used: br = broad signal, s= singlet, d = doublet, t = triplet, dd = double doublet, q = quartet, and m = multiplet. Mass spectra were obtained by a DART-TOF mass spectrometer AccuTOF JMS-T100LC. IR spectra were obtained with a Bruker TENSOR 27 spectrophotometer and a Thermo Fisher Scientific FT-IR NICOLET IS-50. Suitable X-ray quality crystals of **3e**, **3q** compounds were grown through slow evaporation of acetone solution, layered with *n*-hexane. A single white crystal of compounds **3e**, **3q** were mounted on a glass fiber at room temperature. The crystal was then placed on a Bruker SMART APEX CCD diffractometer, equipped with MoKα radiation; decay was negligible in all cases. Systematic absences and intensity statistics were used in the space group determinations. The structures were determined using direct methods.[19] Anisotropic structure refinements were achieved using full-matrix, least-squares techniques on all non-hydrogen atoms. All hydrogen atoms were placed in idealized positions based on hybridization, with isotropic thermal parameters fixed at 1.2 times the value of the attached atom. Structure solutions and refinements were performed using SHELXTL v 6.10.[20] The acetylenic ketones **1** and the bis(trimethylsilyl)ketene acetals **2** were prepared according to published methods.[21,22]

General Procedure for the Synthesis of Tetrasubstituted Allenes 3. To a solution of ynone **1** (0.5 mmol, 1 equiv) and AuCl (0.025 mmol, 5 mol %) in anhydrous 15 mL of CH_2Cl_2 (0.03 M) was added $BF_3{\cdot}Et_2O$ (0.5 mmol, 1 equiv) at 0 °C under N_2 atmosphere, the mixture was stirred for 3 min. Then bis(trimethylsilyl)ketene acetal **2** (1.1 mmol, 2.2 equiv) was added dropwise. The reaction was stirred for 15 min at 0 °C and monitored by TLC. After completion, the reaction mixture was washed with water (10 mL). The organic layer was dried over anhydrous Na_2SO_4, and the solvent was evaporated under vacuum. The product was purified by column chromatography on silica gel using a mixture of hexane and ethyl acetate (88:12 and 85:15) as eluent.

Compound **3a**: 3-(4-methoxyphenyl)-2,2,6,6-tetramethyl-5-phenylhepta-3,4-dienedioic acid. White powder (175 mg, 70%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 85:5), mp 200−202 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 7.36−7.26 (m, 5H), 7.19 (d, J = 8.81 Hz, 2H), 6.86 (d, J = 8.80 Hz, 2H), 3.60 (s, 3H), 1.42 (s, 12H). $^{13}C\{^{1}H\}$ NMR (75 MHz, DMSO-d_6, ppm): δ 203.3, 178.0, 177.9, 158.2, 135.2, 128.3, 128.0, 127.0, 126.8, 114.5, 113.8, 115.4, 113.2, 113.8, 55.0, 45.5, 44.4, 25.6 (2C), 25.5 (2C). IR (ATR, cm^{-1}): ν 2980 (−OH), 1699 (C═O). MS (DART) m/z: 395 $[M + H]^+$. HRMS (TOF) m/z: $[M + H]^+$ calcd for $C_{24}H_{27}O_5$ 395.1858; found 395.1846.

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Compound **3b**: 1,1′-(1-(4-methoxyphenyl)-3-phenylpropa-1,2-diene-1,3-diyl)bis(cyclobutane-1-carboxylic acid). White powder (167 mg, 63%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 85:5), mp 180−182 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.50 (br, 2H), 7.34−7.18 (m, 7H), 6.91 (d, J = 8.8 Hz, 2H), 3.73 (s, 3H), 2.70−2.50 (m, 4H), 2.39−2.28 (m, 2H), 2.24−2.14 (m, 2H), 1.93−1.72 (m, 4H). ^{13}C{^{1}H} NMR (75 MHz, DMSO-d_6, ppm): δ 204.5, 177.2 (2C), 159.0, 134.7, 129.1 (2C), 127.7 (2C), 127.5, 126.5 (2C), 126.4, 115.2, 115.0, 114.7 (2C), 55.6, 49.1 (2C), 31.1 (2C), 31.0 (2C), 16.4 (2C). IR (ATR, cm^{-1}): ν 2932 (−OH), 1639 (C═O). MS (DART) m/z: 419 [M + H]$^+$. HRMS (TOF) m/z: [M + H]$^+$ calcd for $C_{26}H_{27}O_5$ 419.1858; found 419.1865.

Compound **3c**: 1,1′-(1-(4-methoxyphenyl)-3-phenylpropa-1,2-diene-1,3-diyl)bis(cyclopentane-1-carboxylic acid). White powder (196 mg, 69%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 88:12), mp 180−182 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.45 (br, 2H), 7.39 (d, J = 8.84 Hz, 2H), 7.34−7.30 (m, 4H), 7.24−7.19 (m, 1H), 6.92 (d, J = 8.87 Hz, 2H), 3.73 (s, 3H), 2.34−2.17 (m, 4H), 2.09−1.93 (m, 4H), 1.64−1.49 (m, 9H). ^{13}C{^{1}H} NMR (75 MHz, DMSO-d_6, ppm): δ 204.1, 178.2 (2C), 158.7, 135.9, 128.9 (2C), 128.2 (2C), 127.6, 127.3 (2C), 127.0, 115.2, 115.0, 114.4 (2C), 55.9, 55.9, 55.5, 37.5 (2C), 37.4, 36.5, 36.4, 25.2 (4C). IR (ATR, cm^{-1}): ν 2923 (−OH), 1692 (C═O). MS (DART) m/z: 447 [M + H]$^+$. HRMS (TOF) m/z: [M + H]$^+$ calcd for $C_{28}H_{31}O_5$ 447.2171; found 447.2172.

Compound **3d**: 1,1′-(1-(4-methoxyphenyl)-3-phenylpropa-1,2-diene-1,3-diyl)bis(cyclohexane-1-carboxylic acid). White powder (247 mg, 82%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 88:12), mp 172−174 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.42 (br, 2H), 7.40−7.29 (m, 6H), 7.23−7.18 (m, 1H), 6.92 (d, J = 8.89 Hz, 2H), 3.73 (s, 3H), 2.30−1.52 (m, 20H). ^{13}C{^{1}H} NMR (75 MHz, DMSO-d_6, ppm): δ 204.54, 177.2 (2C), 156.0, 135.7, 129.1 (2C), 127.7 (2C), 127.5, 126.5 (2C), 126.4, 115.2, 115.0, 114.7 (2C), 55.6, 49.1 (2C), 31.1 (4C), 31.0 (4C), 16.4 (2C). IR (ATR, cm^{-1}): ν 3052 (−OH), 1713 (C═O). MS (DART) m/z: 475 [M + H]$^+$. HRMS (TOF) m/z: [M + H]$^+$ calcd for $C_{30}H_{35}O_5$ 475.2484; found 475.2461.

Compound **3e**: 3-(2-methoxyphenyl)-2,2,6,6-tetramethyl-5-phenylhepta-3,4-dienedioic acid. White powder (123 mg, 49%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 85:15), mp 97−99 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.35 (br, 2H), 7.45 (d, J = 7.2 Hz, 2H), 7.34−7.24 (m, 3H), 7.21−7.16 (m, 2H), 7.01 (d, J = 7.5 Hz, 1H), 6.92 (t, J = 7.36 Hz,1H), 3.74 (s, 3H), 1.48 (s, 3H), 1.36 (s, 3H), 1.27 (s, 6H). ^{13}C{^{1}H} NMR (75 MHz, DMSO-d_6, ppm): δ 203.0, 178.6, 178.2, 157.3, 136.0, 130.6, 129.2, 128.6 (2C), 127.3 (2C), 126.9, 125.2, 120.4, 113.2, 111.7, 111.1, 55.5, 46.6, 44.5, 26.7 (2C), 25.4, 25.1. IR (ATR, cm^{-1}): ν 2870 (−OH), 1691 (C═O). MS (DART) m/z: 349 [M−(COOH)$_1$]$^+$, 395 [M + H]$^+$. HRMS (TOF) m/z: [M + H]$^+$ calcd for $C_{24}H_{27}O_5$ 395.1858; found 395.1865.

Compound **3f**: 3-(3-methoxyphenyl)-2,2,6,6-tetramethyl-5-phenylhepta-3,4-dienedioic acid. White powder (158 mg, 63%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 85:15), mp 98−100 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.61 (br, 2H), 7.40−7.21 (m, 6H), 7.00−6.95 (m, 2H), 6.83 (dd, 1H, J = 8 Hz, J = 1.75 Hz), 3.71 (s, 3H), 1.42 (s, 12H). ^{13}C{^{1}H} NMR (75 MHz, DMSO-d_6, ppm): δ 204.3, 178.3, 178.2, 159.2, 136.9, 135.6, 129.9, 128.9 (2C), 127.5 (3C), 119.5, 115.6, 115.4, 113.2, 112.9, 55.4, 45.0, 44.8, 26.8 (2C), 26.2 (2C). IR (ATR, cm^{-1}): ν 2972 (−OH), 1691 (C═O). MS (DART) m/z: 349 [M−(COOH)$_1$]$^+$, 395 [M + H]$^+$. HRMS (TOF) m/z: [M + H]$^+$ calcd for $C_{24}H_{27}O_5$ 395.1858; found 395.1859.

Compound **3g**: 2,2,6,6-tetramethyl-3,5-diphenylhepta-3,4-dienedioic acid. White powder (140 mg, 53%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 85:15), mp 101−102 °C. ^{1}H-NMR (300 MHz, acetone-d$_6$, ppm): δ 7.52−7.48 (m, 4H), 7.35−7.29 (m, 4H), 7.25−7.20 (m, 2H), 1.55 (s, 6H), 1.52 (s, 6H). ^{13}C{^{1}H} NMR (75 MHz, DMSO-d_6, ppm): δ 204.3, 178.3 (2C), 135.7 (2C), 129.0 (4C), 127.5 (6C), 115.5 (2C), 45.0 (2C), 26.8 (2C), 26.3 (2C). IR (ATR, cm^{-1}): ν 3058 (−OH), 1699 (C═O). MS (DART) m/z: 277 [M−(COOH)$_2$]$^+$, 319 [M−(COOH)]$^+$, and 365 [M + H]$^+$. HRMS (TOF) m/z: [M + H]$^+$ calcd for $C_{23}H_{25}O_4$ 365.1753; found 365.1759.

Compound **3h**: 2,2,6,6-tetramethyl-3-phenyl-5-(*p*-tolyl)-hepta-3,4-dienedioic acid. White powder (206 mg, 80%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 85:15), mp 110−112 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.56 (br, 2H), 7.39−7.19 (m, 7H), 7.15 (d, J = 8.14 Hz, 2H), 2.27 (s, 3H), 1.39−1.37 (m, 12H). ^{13}C{^{1}H} NMR (75 MHz, DMSO-d_6, ppm): δ 204.0, 178.3, 178.2, 136.7, 135.8, 132.6, 129.5 (2C), 128.8 (2C), 127.4 (2C), 127.3 (3C), 115.3, 115.2, 45.0, 44.9, 26.7 (2C), 26.2 (2C), 21.0. IR (ATR, cm^{-1}): ν 2976 (−OH), 1693 (C═O). MS (DART) m/z: 291 [M−(COOH)$_2$]$^+$, 333 [M−(COOH)]$^+$, and 379 [M + H]$^+$. HRMS (TOF) m/z: [M + H]$^+$ calcd for $C_{24}H_{27}O_4$ 379.1909; found 379.1909.

Compound **3i**: 3-(4-fluorophenyl)-2,2,6,6-tetramethyl-5-phenylhepta-3,4-dienedioic acid. White powder (120 mg, 47%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 88:12), mp 89−91 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.64 (br, 2H, −COOH), 7.41−7.30 (m, 6H), 7.25−7.12 (m, 3H), 1.41−1.40 (m, 12H). ^{13}C{^{1}H} NMR (75 MHz, DMSO-d_6, ppm) δ 204.1, 178.2, 178.0, 163.2 (d, J^1_{C-F} = 242.4 Hz), 135.5, 132.0 (d, J^4_{C-F} = 3.2 Hz), 129.3 (d, J^3_{C-F} = 8.03 Hz, 2C), 128.9 (2C), 127.5, 127.4 (2C), 115.9, 115.7, 114.5 (d, J^2_{C-F} = 73.1 Hz, 2C), 45.0, 44.9, 26.3, 26.2, 25.8, 25.7. ^{19}F{^{1}H} NMR (283 MHz, DMSO-d_6, ppm): δ −61.00 (s). IR (KBr, cm^{-1}): ν 2980 (−OH), 1702 (C═O). MS (DART) m/z: 383 [M + H]$^+$. HRMS (TOF) m/z: [M + H]$^+$ calcd for $C_{23}H_{24}FO_4$ 383.1659; found 383.1645.

Compound **3j**: 3-(4-chlorophenyl)-2,2,6,6-tetramethyl-5-phenylhepta-3,4-dienedioic acid. White powder (142 mg, 57%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 88:12), mp 106−108 °C. ^{1}H-NMR (300 MHz, DMSO-d_6, ppm): δ 12.68 (br, 2H), 7.42−7.30 (m, 8H), 7.25−7.20 (m, 1H), 1.41−1.39 (m, 12H). ^{13}C{^{1}H} NMR (75 MHz, DMSO-d_6, ppm): δ 204.3, 178.1, 178.0, 135.3, 134.6, 132.1, 129.1 (2C), 128.9 (3C), 127.6 (2C), 127.4 (2C), 115.8, 114.6, 45.0, 44.9, 26.7, 26.6, 26.1, 26.1. IR (ATR, cm^{-1}): ν 3493 (−OH), 1676 (C═O). MS (DART) m/z: 267, 353 [M−(COOH)]$^+$, 399 [M + H]$^+$. HRMS (TOF) m/z: [M + H]$^+$ calcd for $C_{23}H_{24}ClO_4$ 399.1363; found 399.1373.

Compound **3k**: 3-(4-bromophenyl)-2,2,6,6-tetramethyl-5-phenylhepta-3,4-dienedioic acid. White powder (147 mg, 63%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 88:12), mp 104−105 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.71 (br, 2H), 7.54 (d, J = 8.65 Hz, 2H), 7.38−7.21 (m, 7H), 1.41 (s, 12H). ^{13}C{^{1}H} NMR (75

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MHz, DMSO-d_6, ppm): δ 204.3, 178.1, 178.0, 135.2, 135.0, 131.9 (2C), 129.4 (2C), 129.0 (2C), 127.6, 127.4 (2C), 120.7, 115.9, 114.6, 44.9, 44.9, 26.7, 26.6, 26.1 (2C). IR (ATR, cm^{-1}): ν 2936 (−OH), 1695 (C═O). MS (DART) m/z: 443 $[M + H]^+$. HRMS (TOF) m/z: $[M + H]^+$ calcd for $C_{23}H_{24}BrO_4$ 443.0858; found 443.0864.

Compound **3l**: 2,2,6,6-tetramethyl-3-phenyl-5-(4-(trifluoromethyl)phenyl)hepta-3,4-dienedioic acid. White powder (137 mg, 58%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 88:12), mp 112−114 °C. ^{1}H NMR (300 MHz, $CDCl_3$, ppm): δ 12.00 (br, 2H), 7.63−7.55 (m, 4H), 7.45−7.29 (m, 5H), 1.63−1.57 (m, 12H). $^{13}C\{^1H\}$ NMR (75 MHz, $CDCl_3$, ppm) δ 205.2, 183.7 (2C), 138.9, 134.4, 129.4, 129.3 (q, J^2_{C-F} = 32.85 Hz, 2C), 128.8 (2C), 127.7 (2C), 127.6, 125.6 (q, J^3_{CF} = 3.97 Hz, 2C), 124.3 (q, J^1_{C-F} = 271.95 Hz), 115.9, 114.3, 45.8, 45.6, 26.4, 26.1 (2C), 26.0. $^{19}F\{^1H\}$ NMR (283 MHz, DMSO-d_6, ppm): δ −62.60 (s). IR (KBr, cm^{-1}): ν 2917 (−OH), 1698 (C═O). MS (DART) m/z: 434 $[M + H]^+$. HRMS (TOF) m/z: $[M + H]^+$ calcd for $C_{24}H_{25}F_3O_4$ 434.1705; found 434.1703.

Compound **3m**: 2,2,6,6-tetramethyl-3-(naphthalen-2-yl)-5-phenylhepta-3,4-dienedioic acid. White powder (131 mg, 54%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 85:15), mp 120−122 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.72 (br, 2H), 7.87−7.78 (m, 4H), 7.63 (d, J = 7.63 Hz, 1H), 7.51−7.44 (m, 4H), 7.35 (t, J = 7.64 Hz, 2H), 7.24 (t, J = 7.4 Hz, 1H), 1.49 (m, 12H). $^{13}C\{^1H\}$ NMR (75 MHz, DMSO-d_6, ppm): δ 204.6, 178.0, 177.8, 135.2, 133.0, 132.7, 132.0, 128.6 (2C), 127.9, 127.8, 127.5, 127.1 (2C), 126.5, 126.1, 125.9, 125.0, 115.4, 115.2, 45.2 (2C), 26.5, 26.4, 26.0, 25.7. IR (ATR, cm^{-1}): ν 2936 (−OH), 1697 (C═O). MS (DART) m/z: 369 $[M-(COOH)]^+$, 414 $[M + H]^+$. HRMS (TOF) m/z: $[M + H]^+$ calcd for $C_{27}H_{27}O_4$ 415.1909; found 415.1902.

Compound **3n**: 3-(benzo[d][1,3]dioxol-5-yl)-2,2,6,6-tetramethyl-5-phenylhepta-3,4-dienedioic acid. White powder (191 mg, 78%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 85:15), mp 115−117 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.61 (br, 2H), 7.41−7.30 (m, 4H), 7.25−7.19 (m, 1H), 6.96 (m, 2H), 6.95−6.83 (m, 2H), 1.42−1.40 (m, 12H). $^{13}C\{^1H\}$ NMR (75 MHz, DMSO-d_6, ppm): δ 204.0, 178.3, 178.2, 147.9, 146.8, 135.7, 129.4, 128.9 (2C), 127.4 (3C), 120.7, 115.3, 115.2, 108.6, 107.9, 101.5, 45.0, 44.9, 26.8, 26.7, 26.2, 26.0. IR (ATR, cm^{-1}): ν 2896 (−OH), 1690 (C═O). MS (DART) m/z: 321 $[M-(COOH)_2]^+$, 409 $[M + H]^+$. HRMS (TOF) m/z: $[M + H]^+$ Calcd for $C_{24}H_{25}O_6$ 409.1651; Found 409.1648.

Compound **3o**: 2,2,6,6-tetramethyl-3-phenyl-5-(thiophen-3-yl)-4l5-hepta-3,4-dienedioic acid. White powder (178 mg, 68%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 85:15), mp 103−105 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.64 (br, 2H), 7.48 (dd, J = 5.07 Hz, J = 2.82 Hz, 1H), 7.40−7.30 (m, 4H), 7.27−7.23 (m, 2H), 7.18 (dd, J = 1.37, J = 5.07 Hz, 1H), 1.45−1.40 (m, 12H). $^{13}C\{^1H\}$ NMR (75 MHz, DMSO-d_6, ppm): δ 204.7, 178.4, 178.3, 136.1, 135.7, 129.0 (2C), 128.1 (2C), 127.6, 127.6, 126.4, 120.6, 115.5, 112.1, 45.1, 45.0, 26.9 (2C), 26.0 (2C). IR (ATR, cm^{-1}): ν 3196 (−OH), 1702 (C═O). MS (DART) m/z: 327 $[M-(COOH)]^+$,371 $[M + H]^+$. HRMS (TOF) m/z: $[M + H]^+$ calcd for $C_{21}H_{23}O_4S$ 371.1317; found 371.1321.

Compound **3p**: 3-butyl-2,2,6,6-tetramethyl-5-(p-tolyl)hepta-3,4-dienedioic acid. White powder (220 mg, 82%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 88:12), mp 98−101 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.30 (br, 2H), 7.22 (d, J = 8.22 Hz, 2H), 7.10 (d, J = 8.08 Hz, 2H), 2.26 (s, 3H), 1.48−1.24 (m, 18H), 0.81 (t, J = 7.16 Hz, 3H). $^{13}C\{^1H\}$ NMR (75 MHz, DMSO-d_6, ppm): δ 200.0, 178.5, 177.1, 136.0, 133.8, 129.2 (2C), 127.3 (2C), 115.0, 113.8, 45.8, 44.6, 30.0, 28.3, 26.7 (2C), 25.3 (2C), 22.6, 21.0, 14.3. IR (ATR, cm^{-1}): ν 2931 (−OH), 1690 (C═O). MS (DART) m/z: 271 $[M-(COOH)_2]^+$, 359 $[M + H]^+$. HRMS (TOF) m/z: $[M + H]^+$ calcd for $C_{22}H_{31}O_4$ 359.2222; found 359.2214.

Compound **3q**: 3-butyl-5-(4-methoxyphenyl)-2,2,6,6-tetramethyl-4l5-hepta-3,4-dienedioic acid. White powder (145 mg, 56%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 88:12), mp 97−99 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.34 (br, 2H), 7.19 (d, J = 8.85 Hz, 2H), 6.89 (d, J = 8.85 Hz, 2H), 3.73 (s, 3H), 2.10−1.92 (m, 2H), 1.49−1.36 (m, 4H), 1.32−1.27 (m, 12H), 0.81 (t, J = 7.13 Hz, 3H). $^{13}C\{^1H\}$ NMR (75 MHz, DMSO-d_6, ppm): δ 199.8, 178.6, 177.7, 158.3, 128.8, 128.5 (2C), 114.7, 114.1 (2C), 113.7, 55.4, 45.9, 44.6, 30.0, 28.3, 26.7, 26.4, 25.3, 25.1, 22.6, 14.34. IR (KBr, cm^{-1}): ν 2926 (−OH), 1692 (C═O). MS (DART) m/z: 329 $[M-(COOH)]^+$, 375 $[M + H]^+$. HRMS (TOF) m/z: $[M + H]^+$ calcd for $C_{22}H_{31}O_5$ 375.2171; found 375.2181.

Compound **3r**: 1,1′-(1-phenylhepta-1,2-diene-1,3-diyl)bis(cyclopentane-1-carboxylic acid). White powder (203 mg, 64%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 88:12), mp 98−101 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.26 (s, 2H), 7.34−7.25 (m, 4H), 7.18−7.13 (m, 1H), 2.29−1.77 (m, 10H), 1.62−1.22 (m, 12H), 0.78 (t, J = 7.27 Hz, 3H). $^{13}C\{^1H\}$ NMR (75 MHz, DMSO-d_6, ppm): δ 200.3, 178.5, 177.4, 137.1, 128.7 (2C), 127.1 (2), 126.9, 114.7, 113.4, 57.9, 55.7, 37.4, 36.9, 35.4, 35.0, 30.0, 29.6, 25.4, 25.3, 24.7, 24.6, 22.7, 14.4. IR (ATR, cm^{-1}): ν 2928 (−OH), 1695 (C═O). MS (DART) m/z: 351 $[M-(COOH)]^+$, 397 $[M + H]^+$. HRMS (TOF) m/z: $[M + H]^+$ calcd for $C_{25}H_{33}O_4$ 397.2379; found 397.2372.

Compound **3s**: 3-(4-ethynylphenyl)-2,2,6,6-tetramethyl-5-(p-tolyl)-4l5-hepta-3,4-dienedioic acid. White powder (146 mg, 59%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 88:12), mp 131−133 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.62 (br, 2H), 7.45 (d, J = 8.47 Hz, 2H), 7.36 (d, J = 8.30 Hz, 2H), 7.22 (d, J = 8.23 Hz, 2H), 7.15 (d, J = 8.21 Hz, 2H), 4.19 (s, 1H), 2.26 (s, 3H), 1.40−1.37 (m, 12). $^{13}C\{^1H\}$ NMR (75 MHz, DMSO-d_6, ppm): δ 204.7, 178.3, 178.2, 137.0, 136.5, 132.4 (3C), 129.7 (2C), 127.5 (2C), 127.4 (2C), 120.7, 116.0, 115.0, 83.9, 81.8, 45.1, 44.9, 26.4 (2C), 26.2 (2C), 21.1. IR (ATR, cm^{-1}): ν 2932 (−OH), 1695 (C═O). MS (DART) m/z: 357 $[M-COOH]^+$, 403 $[M + H]^+$. HRMS (TOF) m/z: $[M + H]^+$ calcd for $C_{26}H_{27}O_4$ 403.1909; found 403.1903.

Compound **3t**: 3-(4-ethynylphenyl)-5-(4-methoxyphenyl)-2,2,6,6-tetramethylhepta-3,4-dienedioic acid. White powder (166 mg, 69%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 88:12), mp 134−136 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ 12.64 (s, 2H), 7.47 (d, J = 8.27 Hz, 2H), 7.38 (d, J = 8.27 Hz, 2H), 7.30 (d, J = 8.66 Hz, 2H), 6.94 (d, J = 8.77 Hz, 2H), 4.20 (s, 1H), 3.74 (s, 3H), 1.41−1.39 (m, 12H). $^{13}C\{^1H\}$ NMR (75 MHz, DMSO-d_6, ppm): δ 204.5, 178.2, 178.1, 158.8, 136.5, 132.3 (2C), 128.6 (2C), 127.4 8 (2C), 127.2, 120.5, 115.8, 115.6, 114.5 (2C), 83.8, 81.7, 55.6, 45.0, 44.8, 26.7, 26.6, 26.2, 26.1. IR (KBr, cm^{-1}): ν 2928 (−OH), 1697 (C═O). MS (DART) m/z: 331

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$[M-(COOH)_2]^+$, 419 $[M + H]^+$. HRMS (TOF) m/z: $[M + H]^+$ calcd for $C_{26}H_{27}O_5$ 419.1858; found 419.1873.

Compound **3u**: 5,5′-(1,4-phenylene)bis(2,2,6,6-tetramethyl-3-(*p*-tolyl)hepta-3,4-dienedioic acid). White powder (112 mg, 40%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 85:15), mp 115−116 °C. ^{1}H NMR (300 MHz, DMSO-d^6, ppm): δ 12.63 (br, 4H), 7.36 (s, 4H), 7.27 (d, J = 8.2 Hz, 4H), 7.14 (d, J = 8.3 Hz, 4H), 2.27(s, 6H), 1.41 (m, 24H). ^{13}C{^{1}H} NMR (75 MHz, DMSO-d_6, ppm): δ 204.4, 204.3, 178.4 (2C), 178.2 (2C), 136.8 (2C), 134.0 (2C), 132.5 (2C), 129.5 (4C), 127.3 (4C), 127.1 (4C), 116.0 (2C), 115.0 (2C), 45.0 (2C), 44.7 (2C), 26.9, 26.8, 26.7 (2C), 26.4, 26.3, 26.1 (2C), 21.1 (2C). IR (KBr, cm^{-1}): ν 2928 (−OH), 1697 (C═O). MS (DART) m/z: 633 $[M-(COOH)]^+$, 679 $[M + H]^+$. HRMS (TOF) m/z: $[M + H]^+$ calcd for $C_{42}H_{47}O_8$ 679.3271; found 679.3278.

Compound **3x**: 3,5-bis(4-methoxyphenyl)-2,2,6,6-tetramethylhepta-3,4-dienedioic acid. White powder (191 mg, 80%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 85:15), mp 215−217 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ12.54 (s, 2H), 7.35 (d, J = 8.75 Hz, 4H), 6.93 (d, J = 8.71 Hz, 4H), 3.73 (s, 6H), 1.42 (s, 12H). ^{13}C{^{1}H} NMR (75 MHz, DMSO-d_6, ppm): δ 203.5, 178.4 (2C), 158.8 (2C), 128.6 (4C), 127.9 (4C), 114.8 (2C), 114.3 (2C), 55.5 (2C), 45.0 (2C), 26.7 (2C), 26.2 (2C). IR (ATR, cm^{-1}): ν 2897 (−OH), 1701 (C═O). MS (DART) m/z: 425 $[M + H]^+$. HRMS (TOF) m/z: $[M + H]^+$ calcd for $C_{25}H_{29}O_6$ 425.1964; found 425.1954.

Compound **3y**: white powder (173 mg, 71%), purification by flash column chromatography (hexane/ethyl acetate (v/v) = 85:15), mp 210−212 °C. ^{1}H NMR (300 MHz, DMSO-d_6, ppm): δ12.59 (s, 2H), 7.344−7.39 (m, 2H), 7.34 (d, J = 8.51, 2H), 7.18 (t, J = 8.76 Hz, 2H), 6.93 (d, J = 8.78 Hz, 2H), 3.73 (s, 3H), 1.41 (s, 12H). ^{13}C{^{1}H} NMR (75 MHz, DMSO-d_6, ppm) δ 203.8, 178.3, 178.2, 163.2 (d, J^1_{C-F} = 244 Hz), 158.8 (2C), 132.3 (d, J^4_{C-F}= 3.26 Hz), 129.3 (d, J^3_{C-F} = 7.96 Hz, 2C), 128.6 (2C), 127.5, 115.8, 115.6, 115.2 (d, J^2_{C-F} = 72.93 Hz, 2C), 114.4, 55.5, 45.1, 45.0, 26.7, 26.6, 26.1 (2C). ^{19}F{^{1}H} NMR (283 MHz, DMSO-d_6, ppm): δ −115.54 (s). IR (ATR, cm^{-1}): ν 2933 (−OH), 1699 (C═O). MS (DART) m/z: 413 $[M + H]^+$. HRMS (TOF) m/z: $[M + 1]^+$ calcd for $C_{24}H_{26}FO_5$ 413.17643; found 413.17497.

Procedure for the Synthesis of Lactone 4. A 10 mL microwave-transparent process vial was filled with allene **3n** (0.200g, 0.489 mmol), CuI (0.024 mmol, 5 mol %), K_2CO_3 (0.135g, 0.978 mmol, 2 equiv), and DMF (2 mL). The vial was sealed with PEEK snap caps and standard PTFE-coated silicone septa. The reaction mixture was then exposed to microwave heating for 30 min at 100 °C. After irradiation, the device was cooled to room temperature with compressed air for 2 min. Then, a saturated solution of NH_4Cl (5 mL) was added to the reaction mixture and extracted with CH_2Cl_2 (3 × 10 mL). The organic layer was filtered over Celite and anhydrous sodium sulfate. The product was purified by column chromatography on silica gel with a mixture of 97:3 hexane/ethyl acetate.

Microwave irradiation experiments were performed using a Monowave 300 single-mode microwave closed reactor, equipped with a noncontact infrared temperature sensor, a direct pressure-control system for measuring the pressure of the reaction vessel contents, and a compressed-air cooling system.

Compound **4**: (*Z*)-3-(benzo[*d*][1,3]dioxol-5-yl)-3-(4,4-dimethyl-5-oxo-3-phenyldihydrofuran-2(3*H*)-ylidene)-2,2-dimethylpropanoic acid. White powder (160 mg, 80%), mp 115−117 °C. ^{1}H NMR (300 MHz, $CDCl_3$, ppm): δ 9.11 (br, 1H), 7.40−7.36 (m, 2H), 7.34−7.29 (m, 3H), 6.79 (d, J = 7.91, 1H), 6.50 (d, J = 1.67 Hz, 1H), 6.46 (dd, J = 7.92, 1.66 Hz, 1H), 6.03 (d, J = 1.47 Hz, 1H), 6.02 (d, J = 1.46 Hz, 1H), 4.08 (s, 1H), 1.40 (s, 3H), 1.35 (s, 3H), 1.23 (s, 3H), 1.11 (s, 3H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$, ppm): δ 183.0, 181.7, 148.1, 147.8, 147.6, 137.0, 129.8 (2C), 128.4 (2C), 127.5, 126.4, 124.7, 123.3, 109.7, 108.5, 101.3, 49.1, 46.7, 46.1, 23.8, 23.5, 22.9 (2C). IR (ATR, cm^{-1}): ν 2930 (−OH), 1796 (C═O), 1692 (C═O). HRMS (TOF) m/z: $[M + H]^+$ calcd for $C_{24}H_{25}O_6$ 409.1651; found 409.1642.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.3c02550.

Copies of ^{1}H and ^{13}C{1H} NMR spectra, X-ray structures with crystallographic data for **3e** and **3q**, and computational data of DFT calculations (PDF)

Accession Codes

CCDC 2306371−2306372 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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